

Orbase Sudoku — Comprehensive Competitive Analysis

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****Purpose**:** Consolidate all competitive research conducted across the Orbase Sudoku product development cycle — app store presence, gameplay features, replay, scoring, ranking, game history, and the Su-Pu (数谱) concept — into a single strategic reference document.

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1. Executive Summary

The mobile Sudoku market is mature, crowded, and segmented. Over 200 apps compete for the same search terms. Yet the competitive analysis reveals a clear and unoccupied position: ****no app combines teaching-quality hints, calm ad-free design, structured progression, replayable solve records, transparent scoring, and fair ranked competition on both platforms.****

This document synthesizes competitive research across seven dimensions — gameplay, hints, scoring, replay, game history, leaderboards, and app store presence — to demonstrate that Orbase Sudoku's Su-Pu (数谱) framework represents a category-defining innovation that no competitor is positioned to replicate quickly.

****Key finding**:** The Su-Pu concept — a recorded solve that is simultaneously a learning record, a performance record, a collectible artifact, and a shareable achievement — has no equivalent in any Sudoku app. The closest analog exists in a different category entirely (Chess.com's game records). This creates a first-mover window for Orbase to define the standard for recorded Sudoku solves.

2. Market Structure & Segmentation

2.1 Category Overview

The mobile Sudoku market divides into four distinct segments:

Segment	Market Share	Monetization	User Profile	Examples
Mass-Market Free	~60%	Heavy ads, IAP to remove	Casual, ad-tolerant, high churn	Sudoku.com (Easybrain), Sudoku - Classic Puzzle
Premium Indie	~5%	Paid upfront (\$3-7)	Design-sensitive, puzzle enthusiasts	Good Sudoku (Zach Gage)
Utility/Power User	~15%	Free/cheap, some ads	Feature-maximalist, variant lovers	Andoku 3, Sudoku 10000
Casual Game	~20%	IAP, lives, boosters	Mobile gamers, progression-seekers	Sudoku Quest, Sudoku Adventure

2.2 Market Dynamics

Dynamic	Implication for Orbase
Category leader (Sudoku.com) has 100M+ downloads but 30-40% of critical reviews complain about ads	"No ads during play" is a powerful acquisition message
Best teaching app (Good Sudoku) is iOS-only and paid-upfront	Cross-platform freemium with teaching creates accessibility
No app has meaningful replay or solve recording	First-mover advantage in Su-Pu concept
Leaderboards are either absent or speed-dominant	Accuracy-weighted ranking with integrity verification is unique
App store visuals are homogenous (blue grids, feature lists)	Cultural design layer (seal icon, Ink Wash themes) creates visual distinction

2.3 The #1 User Complaint Across the Category

Across all ad-supported Sudoku apps, the dominant negative review sentiment is:

> "Too many ads." / "Ads after every puzzle." / "Full screen video ads."

****No major competitor leads their store presence with an ad-free promise.**** This is Orbace's primary acquisition wedge.

3. Competitor Profiles & SWOT Analysis

3.1 Sudoku.com (Easybrain)

****The category leader. Over 100 million downloads. Sets expectations for the mass market.****

| Dimension | Assessment |

|-----|-----|

| ****Puzzle Quality**** | Good. Solver-validated. Consistent difficulty within broad buckets. Thousands of puzzles. |

| ****Hints**** | Reveal a random cell. No explanation. No technique names. No tiered escalation. |

| ****Board UX**** | Clean, responsive, good highlighting. Notes, undo, auto-check. |

| ****Scoring**** | Not a focus. Time and difficulty tracked but no composite score. No breakdown. |

| ****Replay**** | None. No solve recording. Puzzle completed = puzzle gone. |

| ****Game History**** | None. Aggregate stats only (games played, win rate, streaks, best times). No per-puzzle history. |

| ****Leaderboards**** | None. |

| ****Progression**** | Daily Challenges with seasonal themes. No structured level packs. No awards. |

| ****Monetization**** | Aggressive ads: full-screen video after every puzzle, banner during play. IAP to remove: \$4.99/week or \$29.99/year. |

| ****Platform**** | iOS + Android. |

| ****App Store**** | 4.6★ (iOS, 300K+ ratings). Professional but generic screenshots. Blue grid icon. |

****SWOT Analysis:****

| Strengths | Weaknesses |

|-----|-----|

| Massive brand recognition and install base | Ads are the #1 user complaint |

| Reliable puzzle generation at scale | Hints don't teach — reveal only |

Strong daily challenge engagement	No replay, no game history, no progression
Cross-platform	Timer always visible, creates background anxiety
Seasonal content keeps experience fresh	Score and performance tracking is shallow

Opportunities	Threats
Their ad-frustrated users are Orbace's acquisition target	Could add teaching hints if they chose to invest
Could improve hints and retain users longer	Deep pockets for feature development
—	Market dominance creates high barrier for discovery

3.2 Good Sudoku (Zach Gage)

****The quality ceiling. Apple Design Award finalist. Best-in-class teaching and board feel. iOS only.****

Dimension	Assessment
****Puzzle Quality****	Excellent. Hand-tuned difficulty. Consistent solving experience. Smaller curated library (70,000+).
****Hints****	****Best-in-class.**** Interactive tutorial teaches techniques. Hints show "next logical step" with technique names (Naked Single, Hidden Pair). Animated cell highlights. AI-powered hint support.
****Board UX****	Beautiful. Typography-forward. Satisfying completion animation. Excellent highlighting. Smart note auto-removal. Timer off by default.
****Scoring****	None. Deliberate design choice against competition.
****Replay****	None. Can view static completed board. No playback. No move history.
****Game History****	"Previous Puzzles" section: recent puzzles with date and time. Reverse-chronological list only. No filtering. No multiple attempts. No organization.
****Leaderboards****	None. Philosophical choice.
****Progression****	No level packs. Random puzzles by difficulty. "Daily" section with curated puzzles. No streak tracking.
****Monetization****	\$6.99 one-time purchase. No ads, no IAP.
****Platform****	iOS only.
****App Store****	4.8★ (iOS, 8K+ ratings). Design-forward screenshots. Apple Design Award badge.

****SWOT Analysis:****

| Strengths | Weaknesses |

|-----|-----|

| Best teaching system in the category | iOS only — cedes entire Android market |

| Premium design and UX | Paid upfront — limits audience reach |

| Apple Design Award credibility | No progression system — random play only |

| Technique-aware difficulty and hints | No competitive features |

| Respected in puzzle community | No replay, no game history depth |

| Opportunities | Threats |

|-----|-----|

| — | Could launch on Android (most significant competitive threat) |

| — | Could add progression/replay if they change philosophy |

| — | Their hint system is the benchmark Orbace must match |

3.3 Brainium Sudoku

****The strongest learning + scoring combination in the market. Most comprehensive data capture.****

| Dimension | Assessment |

|-----|-----|

| ****Puzzle Quality**** | Good. Solver-validated. Consistent difficulty. |

| ****Hints**** | ****Strong.**** "Smart Hints" explain the logic behind answers. Teaching-oriented. |

| ****Board UX**** | Functional. Customizable. Accessibility-oriented. Some interaction reliability complaints. |

| ****Scoring**** | ****Comprehensive.**** Per-game scoring with breakdown. Best times, averages, per-difficulty stats. Most stat tracking in category. |

| ****Replay**** | None. No move-by-move playback. Stats show what happened but not how. |

| ****Game History**** | ****Most complete in category.**** "Game History" shows past games with difficulty, time, score, date. Multiple plays = multiple entries. But: flat chronological list. No grouping by puzzle. No filtering. No favorites. |

| ****Leaderboards**** | Global and friend leaderboards. But: aggregate scoreboards, not per-puzzle. Speed-dominant. |

| ****Progression**** | Daily puzzles, achievements, stats. No structured level packs. |

| ****Monetization**** | Ads. IAP to remove. |

| ****Platform**** | iOS + Android. |

| ****App Store**** | 4.7★ (iOS, 50K+ ratings). Busy screenshots. Learning message present but not centered. |

****SWOT Analysis:****

| Strengths | Weaknesses |

|-----|-----|

| Most comprehensive data capture | Data is presented as a ledger, not a library |

| Smart Hints teach logic | Ad friction noted in reviews |

| Strong stats dashboard | No replay — data without playback |

| Cross-platform | Flat history without organization |

| Achievement system | Leaderboards speed-dominant |

| Opportunities | Threats |

|-----|-----|

| If they added replay + organization, they'd be closest to Su-Pu | Could become the "data-driven Sudoku" brand |

| — | Their stat depth is the benchmark to exceed |

| — | Interaction reliability complaints = Orbace quality opportunity |

3.4 Andoku 3

****The power user's choice. Feature-maximalist. Variant-heavy. Android only.****

| Dimension | Assessment |

|-----|-----|

| ****Puzzle Quality**** | Good generation. Supports many variants (Killer, X-Sudoku, Squiggly, etc.). |

| ****Hints**** | None. Reveals cells. No teaching. Assumes player knows Sudoku. |

| ****Board UX**** | Functional, not beautiful. Dense menus. Many settings. Multiple input modes. |

| ****Scoring**** | Points based on difficulty + time. Customizable but opaque defaults. |

| ****Replay**** | None. Move history for undo only. No playback. |

| ****Game History**** | ****Best data model in category.**** Saves each play separately. Supports multiple concurrent games. Can export/import data. Organized by variant type and difficulty. But: UX is developer-oriented. History is buried in menus. |

| ****Leaderboards**** | None. |

| ****Progression**** | No structured progression. Difficulty selector + endless play. |

| ****Monetization**** | Free with ads. One IAP to remove. |

| ****Platform**** | Android only. |

| ****App Store**** | 4.7★ (Android, 50K+ ratings). Technical screenshots. Appeals to power users, intimidates casuals. |

****SWOT Analysis:****

| Strengths | Weaknesses |

|-----|-----|

| Best variant support in category | Android only |

| Strong data model (multiple attempts, export) | UX is developer-oriented, not player-friendly |

| Multiple concurrent games supported | No teaching features |

| Data portability (export/import) | No replay despite having the data |

| Power user trust | Visually dated |

| Opportunities | Threats |

|-----|-----|

| Their data model proves Su-Pu is technically feasible | Owns the "variant Sudoku" segment |

| — | Orbace should not compete on variant quantity |

| — | Their export feature is the benchmark for data portability |

3.5 Cracking the Cryptic

****Niche but influential. YouTube-famous for variant Sudoku. Community-driven.****

| Dimension | Assessment |

|-----|-----|

| ****Puzzle Quality**** | ****Highest quality.**** Hand-crafted by celebrity constructors. Very hard. Not generated. |

| ****Hints**** | Video walkthroughs for every puzzle. Community-driven learning. Not in-app hint system. |

| ****Board UX**** | Clean but utilitarian. Not design-led. Web-based + app hybrid. |

| ****Scoring**** | None. Community tracks solve times informally. |

| ****Replay**** | Browser-based video export (not mobile-native). Community shares solves on YouTube. |

| ****Game History**** | Pack completion tracking. Implicit in which packs are owned/completed. No individual puzzle history. |

| ****Leaderboards**** | Informal. Community-maintained solve times in comments/Discord. Not in-app. |

| ****Progression**** | Puzzle packs. Themed collections. Clear pack ownership. |

| ****Monetization**** | Free base app. Paid puzzle packs (\$2-5 each). |

| ****Platform**** | Web + iOS + Android (limited). |

| ****App Store**** | Niche presence. Community-driven discovery. |

****SWOT Analysis:****

| Strengths | Weaknesses |

|-----|-----|

Highest puzzle quality (hand-crafted)	Not scalable — hand-crafted only
Strong community and YouTube presence	No structured app features (replay, scoring, history)
Puzzle pack monetization works	Web-based, not mobile-native
Proves market for "Sudoku as craft"	Hardcore focus limits audience
Celebrity constructor brand	No casual accessibility

| Opportunities | Threats |

|-----|-----|

Their community behavior proves demand for solve sharing and discussion	Their constructors are the brand — hard to compete on that axis
—	Orbace can serve the same community need with better tools
—	Their video-based solve sharing = latent demand for Su-Pu sharing

3.6 NYT Games (Sudoku)

****The minimalist approach. Streak-focused. Daily habit, not skill development.****

| Dimension | Assessment |

|-----|-----|

****Puzzle Quality****	Good. Curated daily puzzles. Consistent.
****Hints****	Basic. No teaching.
****Board UX****	Clean, minimal. Timer. Notes.
****Scoring****	None. Just solved/not solved.
****Replay****	None.
****Game History****	None browsable. Today's puzzle status only. Archive paywalled.
****Leaderboards****	None.
****Progression****	Streak tracking. That's it.
****Monetization****	Subscription for archive access.
****Platform****	iOS + Android. Account-based sync.
****App Store****	Part of NYT Games bundle.

****SWOT Analysis:****

| Strengths | Weaknesses |

|-----|-----|

Strong brand (NYT)	Minimal features beyond daily puzzle
Account-based cloud sync	No learning, no progression, no history
Solve sharing (colored grid) is viral	History is paywalled
Daily habit is powerful retention	No competitive features

| Opportunities | Threats |

|-----|-----|

| Their solve sharing proves social demand | NYT brand is hard to compete with for casual daily players |

| — | Orbace serves a different need (learning + mastery) |

3.7 Chess.com (Cross-Category Analog)

****Not a Sudoku app, but the benchmark for recorded game culture. Included because their game record system is the proof-of-concept for Su-Pu.****

| Dimension | Assessment |

|-----|-----|

| ****Game Records**** | ****Best-in-class.**** Every game saved as PGN (Portable Game Notation). Millions of games recorded daily. |

| ****Replay**** | Full move-by-move replay with analysis. Speed control. Position search. |

| ****Comparison**** | Engine analysis comparison. Side-by-side game comparison. |

| ****Sharing**** | Shareable game links. Embeddable replay viewer. |

| ****Annotations**** | Game annotations. Move comments. Personal notes. |

| ****Technique Profile**** | Opening repertoire. Tactical motif tracking. Accuracy score per game. |

| ****Leaderboards**** | Puzzle Rush leaderboards. Per-puzzle ratings. |

| ****Collection**** | Game archive with filters by date, opponent, opening, result. |

****SWOT Analysis:****

| Strengths | Weaknesses |

|-----|-----|

| Game records are the core of the platform | Not in Sudoku category |

| Every game is a replayable, shareable artifact | — |

| Records create data moat (millions of games) | — |

| Community built around game sharing and analysis | — |

| Opportunities | Threats |

|-----|-----|

| Their model proves Su-Pu works at scale | Could enter Sudoku market |

| Chess culture treats games as artifacts — Sudoku can too | — |

4. Feature-by-Feature Competitive Matrix

4.1 Core Gameplay & Hints

| Feature | Sudoku.com | Good Sudoku | Brainium | Andoku 3 | Cracking the Cryptic
| NYT Games | **Orbace Target** |

|-----|-----|-----|-----|-----|-----|-----|
-----|-----|

| Puzzle generation | Generated | Curated + generated | Generated | Generated |
Hand-crafted | Curated | Generated + solver-validated |

| Unique solution guarantee |  |  |  |  |  |  |

| Difficulty rating | Broad buckets | Technique-aware | Standard | Basic |

Constructor-rated | Standard | Solver-rated with technique scoring |

| Three-tier hints |  |  |  |  |  |  |  |

| Technique names in hints |  |  |  |  |  |  |  (EN + ZH) |

| Technique practice mode |  |  |  |  |  |  | Post-MVP |

| Tutorial | Basic | Interactive | Basic | None | Video | None | Guided solve with
technique instruction |

4.2 Replay & Solve Recording

| Feature | Sudoku.com | Good Sudoku | Brainium | Andoku 3 | Cracking the Cryptic
| NYT Games | Chess.com | **Orbace Target** |

|-----|-----|-----|-----|-----|-----|-----|
-----|-----|-----|

| Save full move history |  |  |  |  |  |  |  |  |

| Replay solve playback |  |  |  |  |  |  |  |  (复盘) |

| Speed control |  |  |  |  |  |  |  |  |

| Step list with labels |  |  |  |  |  |  |  |  (technique labels) |

| Player vs. optimal comparison |  |  |  |  |  |  |  |  (对谱) |

| Side-by-side comparison |  |  |  |  |  |  |  |  (对谱) |

| Solve as artifact/record |  |  |  |  |  |  |  |  |  (Su-Pu/数谱) |

| Share replayable solve |  |  |  |  |  |  |  |  | Post-MVP (传谱) |

| Personal annotations |  |  |  |  |  |  |  |  |  (谱评) |

****Key finding**:** No Sudoku app has any replay capability. Chess.com has all of it.
Orbace can be first to bring this to Sudoku.

4.3 Scoring Systems

Feature	Sudoku.com	Good Sudoku	Brainium	Andoku 3	Cracking the Cryptic	NYT Games	**Orbace Target**
Composite score	✗	✗	✓	✓	✗	✗	✓
Transparent breakdown	✗	N/A	Partial	✗	N/A	N/A	✓ (full breakdown)
Accuracy-weighted	✗	N/A	✗	✗	N/A	N/A	✓ (difficulty-dominant)
Speed as optional bonus	✗	N/A	✗	✗	N/A	N/A	✓ (timer off = no penalty)
Technique-based scoring	✗	✗	✗	✗	✗	✗	✓ (efficiency ratio)
Score classes (official/practice)	✗	N/A	✗	✗	N/A	N/A	✓ (正谱/习谱/重修谱)
Score certificate	✗	N/A	✗	✗	N/A	N/A	✓ (成绩单)
Ranked eligibility flags	✗	N/A	✗	✗	N/A	N/A	✓

****Key finding**:** Existing scoring is either absent (Good Sudoku, NYT) or speed-dominant (Brainium, Andoku). No app has accuracy-weighted, transparent scoring with ranked eligibility classes.

4.4 Game Progress & Play History

Feature	Sudoku.com	Good Sudoku	Brainium	Andoku 3	Cracking the Cryptic	NYT Games	**Orbace Target**
Auto-save current game	✓	✓	✓	✓	✓	✓	✓
Completion history	✗	✓ (recent only)	✓ (full list)	✓ (buried)	✓ (pack-based)	✗	✓ (藏谱阁)
Multiple attempts per puzzle	✗	✗	Partial	✓	✗	✗	✓ (Su-Pu versioning)
Attempts grouped by puzzle	✗	✗	✗	✗	✗	✗	✓
Per-puzzle best tracking	✗	✗	✗	✗	✗	✗	✓ (最佳正谱/最佳)
Improvement deltas	✗	✗	✗	✗	✗	✗	✓ (+1,600 vs v1)
Filterable history	✗	✗	✗	Partial	✗	✗	✓ (difficulty, date, class)
Favorites/bookmarks	✗	✗	✗	✗	✗	✗	✓ (珍藏谱)
Collection statistics	✗	✗	✗	✗	✗	✗	✓ (total, clean, by difficulty)
Data export	✗	✗	✗	✓	✗	✗	Post-MVP
Cloud sync	✗	✗	✗	✗	✓	✓	Phase 3
History as library/collection	✗	✗	✗	✗	✗	✗	✓ (藏谱阁)

****Key finding**:** Brainium captures the most data. Andoku has the best data model. But no app presents history as a browsable, personal collection. No app treats game history as content.

4.5 Leaderboards & Ranking

Feature	Sudoku.com	Good Sudoku	Brainium	Andoku 3	Cracking the Cryptic	NYT Games	**Orbace Target**
Global leaderboards	✗	✗	✓	✗	✗	✗	Phase 3 (世界名谱榜)
Per-puzzle ranking	✗	✗	✗	✗	✗	✗	✓ (题名谱榜)
Daily challenge ranking	✗	✗	✗	✗	✗	✗	✓ (日名谱榜)
Friend leaderboards	✗	✗	✓	✗	✗	✗	✓ (友名谱榜)
Ranked integrity (anti-cheat)	✗	N/A	✗	N/A	N/A	N/A	✓ (multi-layer)
Assist disqualification	✗	N/A	✗	N/A	N/A	N/A	✓ (正谱 only)
Replay verification	✗	N/A	✗	N/A	N/A	N/A	✓ (replay hash)
Leaderboard entries replayable	✗	N/A	✗	N/A	N/A	N/A	✓ (名谱 are Su-Pu)
Seasonal rankings	✗	N/A	✗	N/A	N/A	N/A	✓ (季名谱榜)
Platform-native (GC/GP)	✗	N/A	✗	N/A	N/A	N/A	Phase 2

****Key finding**:** Leaderboards in Sudoku are rare and, where they exist, speed-dominant and integrity-weak. No app has per-puzzle ranking with replay verification and assist disqualification.

4.6 App Store Presence

Feature	Sudoku.com	Good Sudoku	Brainium	Andoku 3	**Orbace Target**
Icon distinctiveness	Low (blue grid)	Medium (abstract)	Low (blue grid)	Low (number grid)	**High (vermilion seal)**
Screenshot lead message	"Classic sudoku"	Design-forward	"Most popular"	Feature dump	**"No ads. Just focus."**
Ad-free promise visible	✗	Implicit (paid)	✗	✗	**Screenshot 1**
Cultural visual identity	None	None	None	None	**Ink Wash, seal, Tea Moment**
Rating	4.6	4.8	4.7	4.7	**Target: 4.7+**
Review sentiment risk	"Too many ads"	"iOS only"	"Ads" / "Bugs"	"Looks dated"	**Ad-free removes #1 complaint**

5. Deep Dive: Game Progress & Play History

5.1 How Each Competitor Treats Player History

App	Philosophy	Player's Relationship to History
Sudoku.com	"Keep playing. Don't look back."	History doesn't matter. Just aggregate stats.
Good Sudoku	"Here's what you've done recently."	History is a log. Glanceable, not explorable.
Brainium	"We track everything. Here's the data."	History is a spreadsheet. Comprehensive, not warm.
Andoku 3	"All data is saved. Find it if you can."	History is a file system. Powerful, not designed.
NYT Games	"Did you do today's puzzle?"	History is a streak counter. Habit, not journey.
Cracking the Cryptic	"Complete the pack."	History is pack progress. Collection, not reflection.

5.2 What Everyone Does (Table Stakes)

- Auto-save current game state: 100%
- Resume from home screen: 100%
- Basic stats (games played, best times): 90%
- Streak tracking: 70%

5.3 What Almost No One Does (Opportunity Gaps)

Capability	Adoption	Orbase Implementation
Browseable completion history	20%	藏谱阁 — full Su-Pu library
Multiple attempts per puzzle	10%	Su-Pu versioning (v1, v2, v3)
Per-puzzle detail view	10%	Full Su-Pu detail with score, techniques, notes
History filterable by difficulty/date/class	0%	All, 正谱, 净谱, 珍藏谱, 残谱 filters
Favorites/bookmarks	0%	珍藏谱
Improvement deltas between attempts	0%	"+1,600 vs v1" / "+4 steps faster"
Collection statistics	0%	Total, clean, by difficulty, PB rate
History as emotional library	0%	"Your Sudoku journey" framing

5.4 The Core Insight

Every Sudoku app treats play history as **metadata** — a byproduct that gets summarized into aggregate stats. No app treats play history as **content** — a personal library the player wants to browse, revisit, and feel ownership over.

Orbase's Su-Pu concept transforms history from metadata into content. The 藏谱阁 (Record Hall) is not a history screen. It is the player's Sudoku body of work.

6. Deep Dive: Replay & Solve Recording

6.1 The Current State

****Replay does not exist in any mobile Sudoku app.****

App	Replay?	Closest Feature
-----	-----	-----
Sudoku.com		None
Good Sudoku		Static completed board view
Brainium		Stats show what happened, not how
Andoku 3		Move history for undo only
Cracking the Cryptic		Video export (browser-based, not mobile)
NYT Games		None

6.2 Why Replay is Absent

1. ****Data architecture****: Most apps store only final board state. Move-level recording requires forethought.
2. ****Perceived low demand****: Mass-market apps optimize for casual "complete and move on" behavior.
3. ****No competitive pressure****: Since no one has it, no one has been forced to build it.

6.3 The Chess.com Proof of Concept

Chess.com's game records (PGN format) demonstrate that recorded solves work at massive scale:

- Millions of games recorded daily
- Each game is replayable, shareable, analyzable
- Players build collections of their games
- Famous games are studied
- Game records are the platform's core content

****Sudoku has the same potential. No one has recognized it.****

6.4 Orbace's Replay System (复盘)

Capability	Market Status	Orbace Implementation
-----	-----	-----

Full move-by-move playback	None	复盘 viewer with speed control
Step list with technique labels	None	Each step labeled (NS, HS, NP, etc.)
Player vs. optimal comparison	None	Side-by-side synchronized 对谱
Divergence detection	None	Auto-identify where paths split
Replay without creating new attempt	None	Viewing is consequence-free
Retry with version tracking	None	重解 creates new Su-Pu with parent link

7. Deep Dive: Scoring Systems

7.1 How Competitors Score

App	Scoring Approach	Strengths	Weaknesses
Sudoku.com	Time + difficulty tracked. No composite score.	Simple	Not a scoring system
Good Sudoku	**No scoring.** Philosophical choice.	Avoids competition anxiety	No performance measurement
Brainium	Points based on difficulty + time + errors	Most comprehensive in category	Opaque formula. Speed-dominant.
Andoku 3	Points based on difficulty + time	Customizable	Confusing defaults
NYT Games	None. Solved/not solved.	Simple	No measurement

7.2 The Speed Problem

All existing scoring systems are speed-dominant. This creates two problems:

1. **Incentivizes rushing**: Players are rewarded for speed, not accuracy or elegance
2. **Contradicts calm**: Timer pressure is the opposite of the calm experience many players want

7.3 Orbase's Scoring Differentiation

Principle	Implementation
Accuracy-dominant	Errors and hints reduce a multiplier. Difficulty determines base.
Speed is optional	Time bonus only when timer enabled. Capped at +10%.
Transparent	Full breakdown shown: base, accuracy multiplier, each penalty, each bonus
Class-based	正谱 (official), 习谱 (practice), 重修谱 (retry), 旧谱 (legacy)
Deterministic	Same puzzle + same performance = same score

| **Versioned** | Scoring version stored with every Su-Pu |
| **Efficiency-aware** | Step count vs. optimal path tracked |

8. Deep Dive: Leaderboards & Ranking

8.1 How Competitors Handle Ranking

App	Leaderboards?	Type	Fairness
Sudoku.com	✗	None	N/A
Good Sudoku	✗	None (philosophical)	N/A
Brainium	✓	Global + friend	Aggregate scores. Speed-dominant. No per-puzzle ranking. No assist disqualification.
Andoku 3	✗	None	N/A
NYT Games	✗	None	N/A

8.2 The Integrity Gap

Where leaderboards exist (Brainium), they suffer from:

- Aggregate scoring across different puzzles (unfair comparison)
- Speed-dominant ranking (rewards rushing)
- No assist disqualification (hinted solves compete with clean solves)
- No verification (scores could be manipulated)

8.3 Orbase's Leaderboard Differentiation

Principle	Implementation
Per-puzzle only	Same puzzle for all players on a leaderboard
Su-Pu based	Every entry is a replayable, verifiable Su-Pu
Assist-disqualified	Only 正谱 class Su-Pu ranked
Integrity-verified	Multi-layer validation: replay hash, puzzle checksum, time bounds, score bounds, move sequence
Opt-in competitive	Behind Extreme unlock (Scholar's Path Stage 3)
Calm presentation	No rank-loss anxiety. No push notifications unless opted in.
Replayable entries	Every leaderboard entry can be replayed and compared
Phased rollout	Local → Platform-native → Worldwide

9. Deep Dive: App Store Presence

9.1 Visual Patterns in the Category

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Icon strategy:

- 80% → Number grid on blue/purple gradient (undifferentiated)
- 10% → Stylized abstract (Good Sudoku)
- 10% → Generic puzzle piece / number

Screenshot strategy:

- 90% → Feature list: board, daily, hints, stats, themes
- 5% → Design-led editorial (Good Sudoku)
- 5% → Technical feature dump (Andoku)

Lead message:

- 60% → "Thousands of puzzles" (volume)
- 20% → "Free Sudoku" (price)
- 15% → "Classic Sudoku" (category)
- 5% → "Learn/improve" (value)

...

9.2 The Three Store Gaps

Gap	Description	Orbase Opportunity
No app leads with "calm" visually	Every icon uses bright colors and gradients	Vermilion seal on rice paper is unique and signals calm
No app leads with ad-free promise	Despite "too many ads" being #1 complaint	Screenshot 1: "No ads. Just focus."
No app has cultural visual identity	All use Western design language	Ink Wash, seal, Tea Moment, Scholar's Path

9.3 Review Sentiment Analysis

Positive Themes	Negative Themes
"Great puzzles"	"Too many ads" (#1 complaint across category)
"Good difficulty"	"Ads after every puzzle"
"Actually teaches" (Good Sudoku/Brainium only)	"Can't stand the interruptions"
"Beautiful design" (Good Sudoku)	"Not on Android" (Good Sudoku)
	"Looks dated" (Andoku)

| | "Recent update broke things" (Brainium) |

10. The Su-Pu White Space

10.1 What Su-Pu Combines

No competitor combines these elements. Fragments exist across different apps:

Fragment	Who Has It	What's Missing
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Per-game data saved	Brainium, Andoku	No replay, no organization, no sharing
Technique awareness	Good Sudoku	Aggregate only, not per-solve, no replay
Multiple attempts	Andoku	Not grouped by puzzle, no comparison
Solve sharing	NYT Games	Static result image, not replayable
Community solve culture	Cracking the Cryptic	External (YouTube/Discord), not in-app
Data portability	Andoku	File export only, no sharing UX
Game as artifact	Chess.com (not Sudoku)	Full system exists but in wrong category

10.2 The Chess.com Parallel

Chess.com Feature	Su-Pu Equivalent
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PGN game records	Su-Pu (数谱)
Game Review	复盘 (Replay)
Engine analysis	Optimal path comparison
Game archive	藏谱阁 (Record Hall)
Shareable games	传谱 (Share)
Puzzle Rush leaderboards	名谱榜 (Per-puzzle leaderboards)
Opening explorer	Technique explorer

10.3 Why No One Has Built This

Reason	Explanation
-----	-----
Architecture debt	Most apps built years ago. Move-level recording requires rearchitecture.
Mass-market focus	Dominant apps optimize for casual "complete and move on."
No competitive pressure	Since no one has replay, no one has been forced to build comparison or sharing.

| Category blindness | Sudoku seen as "casual puzzle." Developers don't look to Chess.com for inspiration. |
| Philosophical choice | Good Sudoku deliberately avoids competitive features and persistent records. |

10.4 The First-Mover Window

The window is open. No competitor has announced replay or Su-Pu-like features. Chess.com hasn't entered Sudoku. Good Sudoku is philosophically opposed. Sudoku.com is structurally committed to their ad model.

****Orbace can define the category for recorded Sudoku solves.****

11. Orbace Differentiation Strategy

11.1 Positioning Statement

> Orbace Sudoku is the calm app that teaches you to solve better — with every solve preserved as a Su-Pu you can replay, compare, share, and improve — and for those who seek mastery, a fair arena to prove it.

11.2 Differentiation Pillars

Pillar	Competitive Contrast	MVP Priority
****Su-Pu (Recorded Solves)****	No app has anything similar	High
****Teaching Hints****	Only Good Sudoku matches. iOS-only, paid.	High
****Calm by Design****	#1 complaint in category is ads. Orbace: zero during play.	High
****Transparent Scoring****	Existing scoring is absent or speed-dominant	High
****Cultural Identity****	All apps use Western design. Orbace: Ink Wash, seals, Tea Moment.	High
****Staged Awards****	No app gates content behind skill	High
****Fair Ranking****	Where leaderboards exist, they lack integrity	Medium
****Cross-Platform****	Good Sudoku iOS-only. Andoku Android-only.	High

11.3 Competitive Moat

Moat Layer	Mechanism	Time to Replicate
****Data accumulation****	Players build Su-Pu collections. Competitor can't import.	
Years (data is non-transferable)		

Network effects	Su-Pu sharing and 名谱榜 create community gravity	Years (requires critical mass)
Brand ownership	Su-Pu/数谱 terminology + cultural identity	Permanent (trademark + culture)
Feature depth	Replay + comparison + versioning + sharing + ranking	12-18 months (full rebuild)
Trust	Ranked integrity requires proven track record	Years (reputation)

11.4 What Orbase Does NOT Compete On

- ~~✗~~ Puzzle volume ("thousands of puzzles" — everyone claims this)
- ~~✗~~ Variant quantity (Andoku owns this segment)
- ~~✗~~ Hand-crafted constructor puzzles (Cracking the Cryptic owns this)
- ~~✗~~ Raw speed leaderboards (incentivizes bad solving)
- ~~✗~~ "Free" as primary message (devalues the product)

12. Competitive Risk Register

Risk	Severity	Likelihood	Mitigation
Good Sudoku launches on Android + adds progression	High	Medium	Paid-upfront limits reach. Orbase's freemium + Su-Pu + ranking is a different product.
Sudoku.com adds teaching hints	Medium	Low	Ad model disincentivizes features that extend session time. Teaching requires technique engine they lack.
Brainium adds replay and organization	Medium	Low	Would require rearchitecting their data model. Their UX philosophy is spreadsheet, not library.
Chess.com enters Sudoku	High	Low	They're focused on chess. But if they did, they have the architecture. Monitor.
AI-powered Sudoku app emerges	Medium	Low-Medium	LLMs are unreliable for deterministic logic. Technique-based hints are verifiably correct.
Market dismisses as "another Sudoku app"	High	High	Differentiation must be visible in first 5 seconds: seal icon, ad-free screenshot 1, cultural aesthetic.
"Calm" reads as "boring"	Medium	Medium	Haptics, seal animations, and completion delight deliver emotion without noise.
Cheating undermines leaderboard trust	High	Medium	Multi-layer validation. Move sequence verification. Rapid suspension.
Backend costs exceed budget	Medium	Medium	Phased rollout: local → platform-native → worldwide. Platform-native is free.

13. Strategic Recommendations

13.1 Immediate Actions (MVP)

1. **Lead with "No ads during play" in store presence.** This directly addresses the #1 category complaint and no competitor makes this their lead message.
2. **Ship the vermilion seal icon.** In a sea of blue grid icons, distinctive = noticed. Noticed = tapped.
3. **Match Good Sudoku's hint quality.** Three-tier hints with technique names in English and Chinese. This is the teaching benchmark and must be met.
4. **Build Su-Pu from day one.** Full move history capture, auto-save, 藏谱阁 collection. Data architecture is the foundation for every differentiator.
5. **Ship cultural design layer in MVP.** Ink Wash theme, seal completions, Tea Moment daily puzzle, Scholar's Path naming. These are design assets, not feature development.

13.2 Short-Term (Post-MVP, 6 months)

6. **Implement 对谱 comparison.** Side-by-side replay with divergence detection. This is the learning engine.
7. **Launch platform-native leaderboards.** Game Center and Google Play Games. Free infrastructure, immediate competitive credibility.
8. **Build 传谱 sharing.** Shareable Su-Pu with deep links. Enable the community behavior Cracking the Cryptic's audience already demonstrates.
9. **Pursue featuring opportunities.** Apple Design Award submission. Google Play Best Of submission. "First Sudoku app with solve recording" is a compelling story.

13.3 Long-Term (Year 1-2)

10. **Launch worldwide 名谱榜.** Cross-platform backend with integrity verification. Own the "fair competitive Sudoku" position.
11. **Build technique explorer.** Aggregate technique data from millions of Su-Pu. "How players solve" becomes a product in itself.
12. **Enable community Su-Pu exchange.** Public Su-Pu, 谱评 comments, following other players. Network effects create retention.

13. ****Consider Chess.com partnership or acquisition defense.**** If Chess.com enters Sudoku, having the established Su-Pu ecosystem is the best defense.

13.4 The Single Most Important Principle

> ****Every puzzle solved in Orbace creates a Su-Pu. Every Su-Pu lives in the player's 藏谱阁. Every Su-Pu can be replayed, compared, annotated, shared, and improved.****

This is the organizing principle that no competitor has. It transforms Sudoku from disposable consumption into a personal craft with a growing body of work. It creates data that accumulates, relationships that deepen, and a product that becomes harder to leave the longer you use it.

End of Comprehensive Competitive Analysis